COMICCRAFT — TECHNOLOGY STACK

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BACKEND

- Python 3.10+
- FastAPI (REST API framework)
- Uvicorn (ASGI server)
- Jinja2 (HTML templating)

AI - TEXT GENERATION

- Google Gemini API
- Model: gemini-3.8-flash
- SDK: google-genai
- Used for: 5-panel outline and full narration/dialogue

AI - IMAGE GENERATION

- Pollinations.ai
- Free image generation service, no API key required
- HTTP endpoint: https://image.pollinations.ai/prompt/{prompt}
- Used for: Comic-style panel illustrations

DOCUMENT PROCESSING

- fpdf2 (PDF export)
- Pillow (image handling)
- Requests (HTTP calls to Pollinations)

ENVIRONMENT MANAGEMENT

- python-dotenv (reads GOOGLE_API_KEY from .env)
- Virtual environment (venv)

DEVELOPMENT TOOLS

- VS Code
- Git & GitHub
- Windows PowerShell terminal

WHY THESE CHOICES

FastAPI: Modern, fast, automatic API docs, great async support

Jinja2: Simple HTML templating, no JavaScript framework needed

Gemini Flash: Fast, free tier available, handles creative writing well

Pollinations.ai: Free alternative to Stable Diffusion, no GPU required

fpdf2: Pure-Python PDF generation, easy to bundle in the same app